

RSA Cryptography Exercises

Q&A Notes

Question 1

Consider an RSA key set with

$$p = 11, \quad q = 29, \quad n = 319, \quad e = 3.$$

What value of d should be used in the secret key? What is the encryption of the message $M = 100$?

Answer 1

The secret exponent is

$$d = 187$$

and the encryption of $M = 100$ is

$$254.$$

First compute Euler's totient of $n = pq$:

$$\phi(n) = (p - 1)(q - 1) = (11 - 1)(29 - 1) = 10 \cdot 28 = 280.$$

The secret exponent d must satisfy

$$ed \equiv 1 \pmod{\phi(n)}.$$

Since $e = 3$, we need

$$3d \equiv 1 \pmod{280}.$$

Using Extended Euclid,

$$280 = 3(93) + 1,$$

so

$$1 = 280 - 3(93).$$

Thus

$$-93$$

is an inverse of 3 modulo 280. Reduce it:

$$-93 \equiv 187 \pmod{280}.$$

Therefore

$$d = 187.$$

Now encrypt $M = 100$ using

$$C \equiv M^e \pmod{n}.$$

So

$$C \equiv 100^3 \pmod{319}.$$

Compute step by step:

$$100^2 = 10000 \equiv 111 \pmod{319},$$

because

$$10000 = 319(31) + 111.$$

Then

$$100^3 \equiv 111 \cdot 100 = 11100 \pmod{319}.$$

Since

$$11100 = 319(34) + 254,$$

we get

$$100^3 \equiv 254 \pmod{319}.$$

Final RSA Values

Public key: $(e, n) = (3, 319)$.

Secret key: $(d, n) = (187, 319)$.

Ciphertext for $M = 100$: $C = 254$.

Question 2

Prove that if Alice's public exponent is $e = 3$ and an adversary obtains Alice's secret exponent d , where $0 < d < \phi(n)$, then the adversary can factor Alice's modulus n in polynomial time in the number of bits in n .

Answer 2

If the adversary knows d , then they can recover $\phi(n)$, and once $\phi(n)$ is known, factoring $n = pq$ is easy.

In RSA, d is chosen so that

$$ed \equiv 1 \pmod{\phi(n)}.$$

Here $e = 3$, so

$$3d \equiv 1 \pmod{\phi(n)}.$$

That means

$$3d - 1 = k\phi(n)$$

for some positive integer k .

Because

$$0 < d < \phi(n),$$

we have

$$0 < 3d - 1 < 3\phi(n).$$

Therefore

$$k \in \{1, 2\}.$$

So the adversary tries the two possible values

$$\phi(n) = 3d - 1$$

or

$$\phi(n) = \frac{3d - 1}{2},$$

keeping the one that is an integer and gives valid factors.

Now suppose $n = pq$. Then

$$\phi(n) = (p - 1)(q - 1) = pq - p - q + 1.$$

Since $pq = n$,

$$\phi(n) = n - (p + q) + 1.$$

Thus

$$p + q = n - \phi(n) + 1.$$

Let

$$s = p + q.$$

Then p and q are the two roots of

$$t^2 - st + n = 0.$$

So

$$t = \frac{s \pm \sqrt{s^2 - 4n}}{2}.$$

The adversary can compute s , compute the discriminant

$$\Delta = s^2 - 4n,$$

take its square root, and recover

$$p = \frac{s + \sqrt{\Delta}}{2}, \quad q = \frac{s - \sqrt{\Delta}}{2}.$$

Why This Is Polynomial Time

Only a constant number of candidates for $\phi(n)$ are tested. Arithmetic operations, integer square root, and checking the factors all take polynomial time in the number of bits of n .

Question 3

Prove that RSA is multiplicative in the sense that

$$P_A(M_1)P_A(M_2) \equiv P_A(M_1M_2) \pmod{n}.$$

Use this fact to prove that if an adversary had a procedure that could efficiently decrypt 1% of messages from $\mathbb{Z}_n$ encrypted with P_A , then the adversary could employ a probabilistic algorithm to decrypt every message encrypted with P_A with high probability.

Answer 3

RSA is multiplicative because encryption is exponentiation modulo n .

Let Alice's public encryption function be

$$P_A(M) \equiv M^e \pmod{n}.$$

Then

$$P_A(M_1)P_A(M_2) \equiv M_1^e M_2^e \pmod{n}.$$

Using the exponent rule,

$$M_1^e M_2^e = (M_1 M_2)^e.$$

Therefore

$$P_A(M_1)P_A(M_2) \equiv (M_1 M_2)^e \equiv P_A(M_1 M_2) \pmod{n}.$$

So

$$P_A(M_1)P_A(M_2) \equiv P_A(M_1 M_2) \pmod{n}.$$

Now suppose an adversary has a procedure D that decrypts 1% of RSA ciphertexts efficiently. The adversary wants to decrypt a target ciphertext

$$C = P_A(M).$$

The adversary chooses a random invertible value

$$r \in \mathbb{Z}_n^*$$

and computes

$$C' = C \cdot P_A(r) \pmod{n}.$$

Using the multiplicative property,

$$C' \equiv P_A(M)P_A(r) \equiv P_A(Mr) \pmod{n}.$$

So C' is an encryption of the randomized message Mr .

If the decryption procedure succeeds on C' , it returns

$$Mr \pmod{n}.$$

Since $r \in \mathbb{Z}_n^*$, the inverse $r^{-1} \pmod{n}$ exists. The adversary then computes

$$M \equiv (Mr)r^{-1} \pmod{n}.$$

Each random choice of r gives the decryption procedure a fresh-looking encrypted message. Since the procedure works on 1% of messages, one trial succeeds with probability 0.01. After T independent trials, the probability of failure is at most

$$(0.99)^T.$$

Thus, by repeating enough times, the adversary decrypts the original message with high probability.

Main Idea

The adversary hides the target message M by multiplying it by a random invertible r . If the weak decryptor can decrypt $P_A(Mr)$, then multiplying by r^{-1} recovers M .